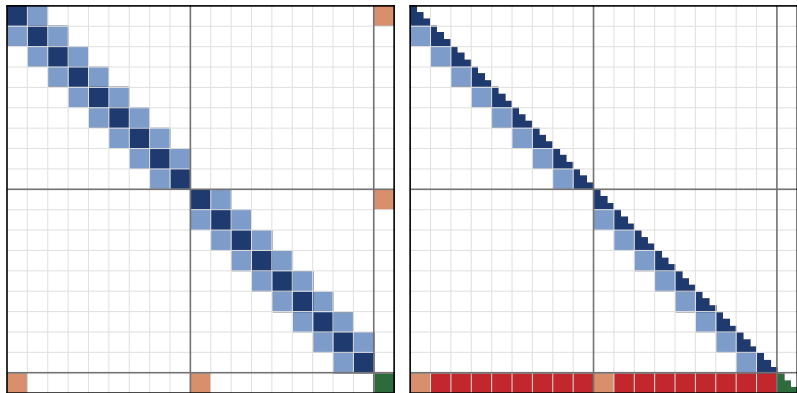


tail diagonal blocks
 root block
 fill-in

tail off-diagonal blocks
 root-tail coupling
 horizon-level parallelism hierarchy

matrix: nnz = 495

factor: nnz = 420, fill-in = 144

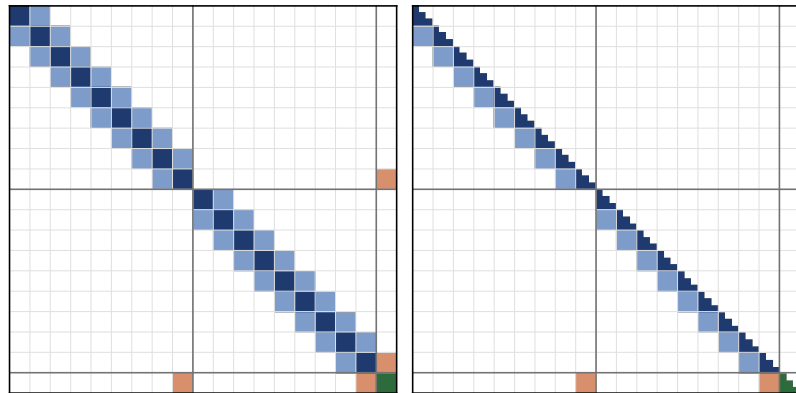


variable order: $(x_1^i, x_2^i, x_3^i, x_4^i, x_5^i, x_6^i, x_7^i, x_8^i, x_9^i)_{i=1,2}, x_0$

(a) forward ordering, sequential in horizon

matrix: nnz = 495

factor: nnz = 276, fill-in = 0

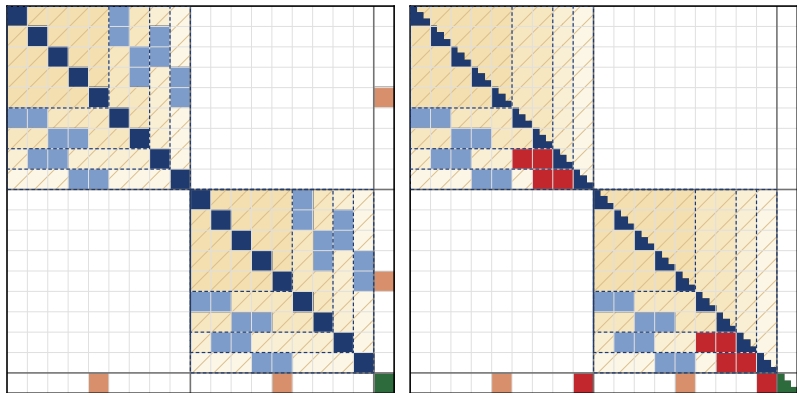


variable order: $(x_9^i, x_8^i, x_7^i, x_6^i, x_5^i, x_4^i, x_3^i, x_2^i, x_1^i)_{i=1,2}, x_0$

(b) reverse ordering, sequential in horizon

matrix: nnz = 495

factor: nnz = 366, fill-in = 90

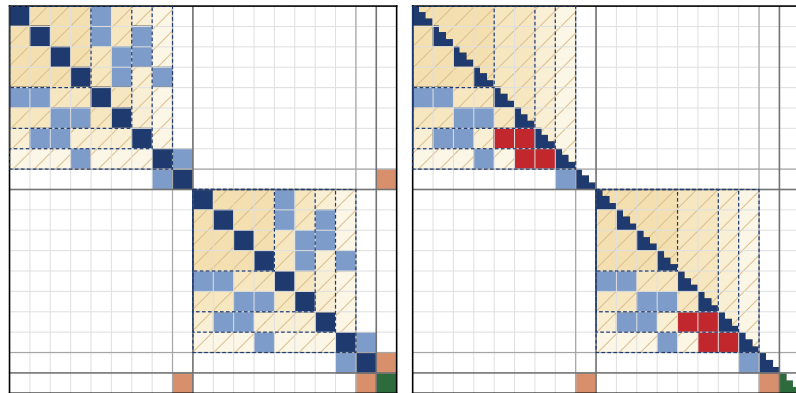


variable order: $(x_9^i, x_7^i, x_5^i, x_3^i, x_1^i, x_8^i, x_4^i, x_6^i, x_2^i)_{i=1,2}, x_0$

(c) reverse ordering, parallel in horizon (x_1^i not separated)

matrix: nnz = 495

factor: nnz = 348, fill-in = 72



variable order: $(x_9^i, x_7^i, x_5^i, x_3^i, x_8^i, x_4^i, x_6^i, x_2^i, x_1^i)_{i=1,2}, x_0$

(d) reverse ordering, parallel in horizon (x_1^i separated)